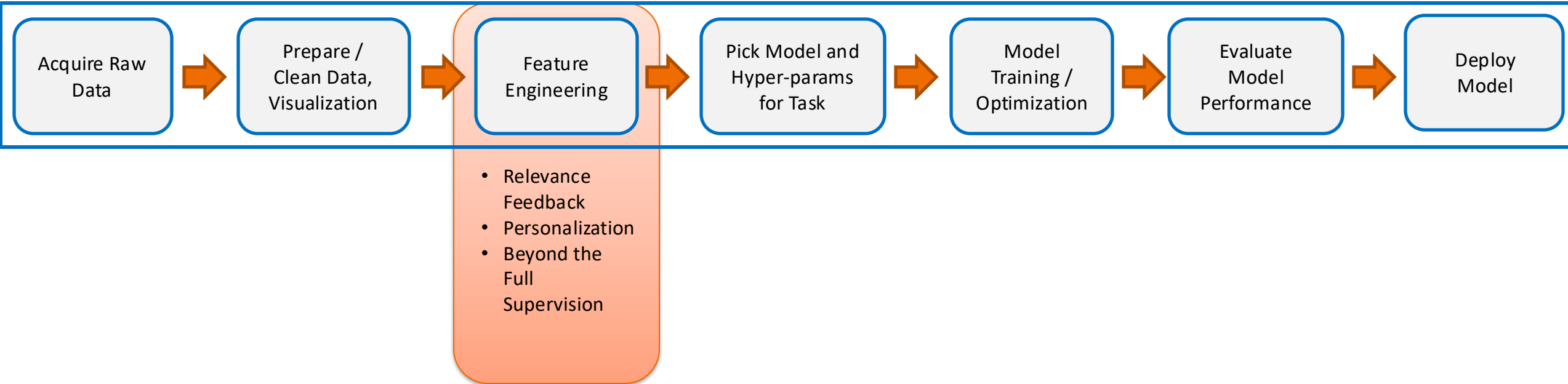
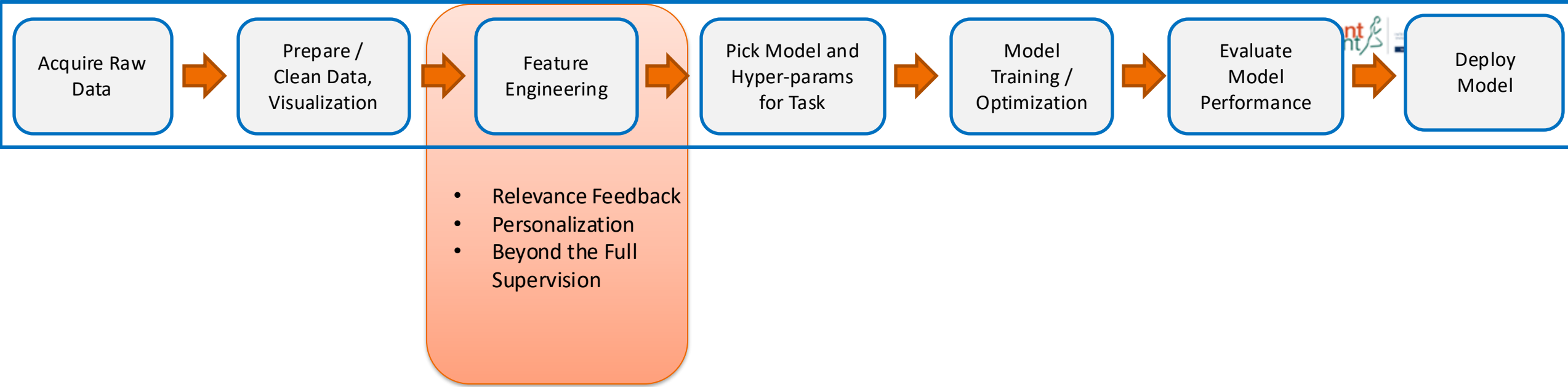


Focus for this lecture



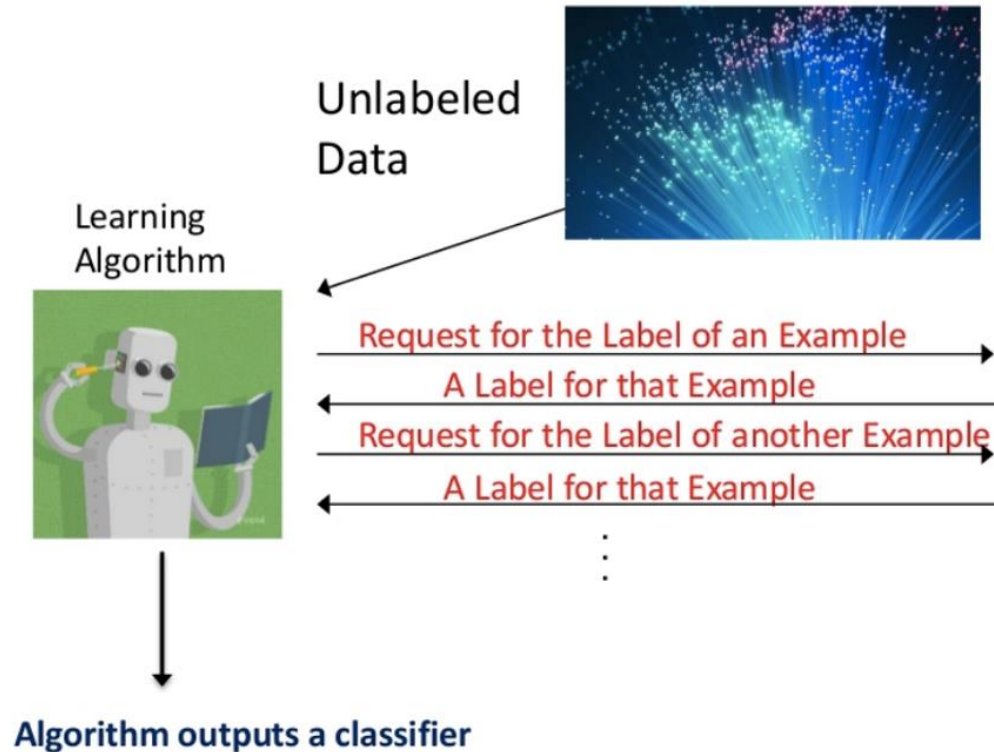


ML Systems with User in the Loop

Different Roles for Human in the Loop

- An expert in the loop during training a system
 - Active Learning
- An expert/human in the loop while deploying the system
 - Simpler tasks are first given to the machine.
- A human co-worker who shares the same task or complement with an ML system. Learns by watching human/expert
 - Incremental Learning, Intelligent Task Division
- A user who interacts with the system
 - Cooperative users, User Feedback; Personalization, incentivize

Learn with minimal # of examples ? Active Learning



Expert / Oracle **Active Learning**



- Stream-Based Active Learning
 - Consider one unlabeled example at a time Decide whether to **query its label** or **ignore it**
- Pool-Based Active Learning
 - Given: a large unlabeled pool of examples **Rank** examples in **order of informativeness**
 - **Query the labels** for the most informative example(s)

**Q: HOW MANY SAMPLES ARE REQUIRED TO
LEARN THE CONCEPT OF RECTANGLES?**

**HOW DO I DEPLOY WHEN THE SYSTEM IS NOT
100% ACCURATE?**

Situation

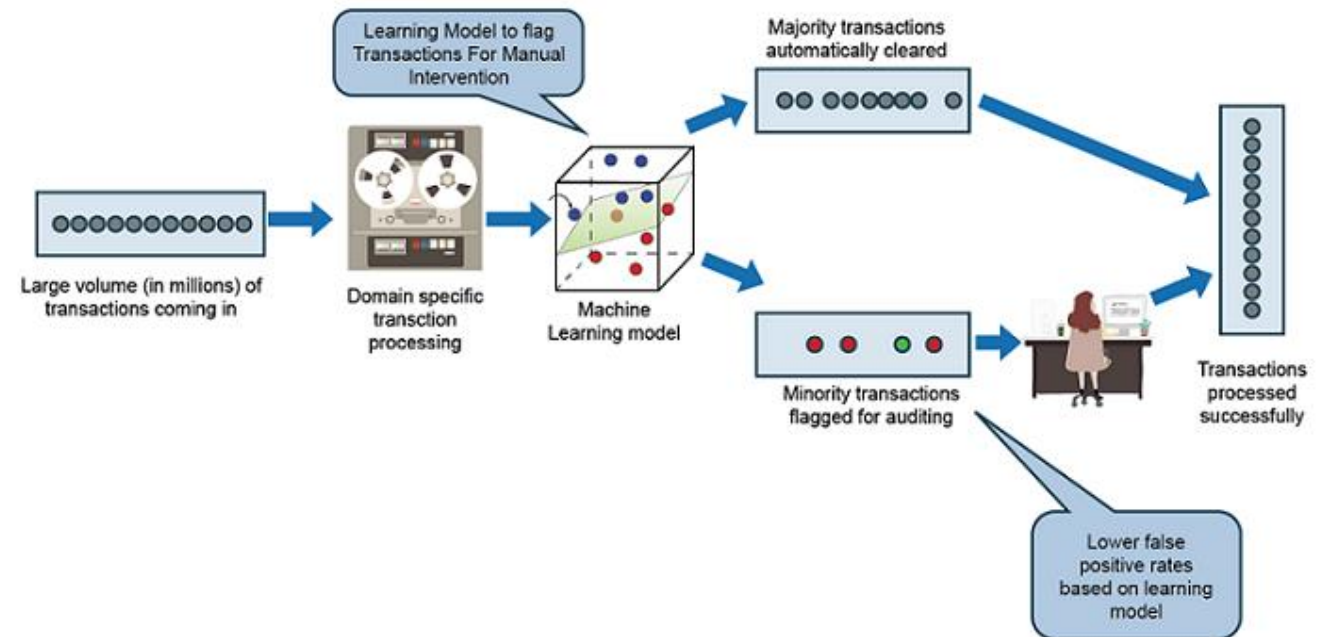
- System does easy tasks
 - System decides what is easy and what is difficult
- Harder tasks are given to humans
 - Human load reduces significantly
- With time
 - Specific hard situations are found and examples gets created
- System improves;
 - Automatically
 - With expert interventions

Users in the ML systems

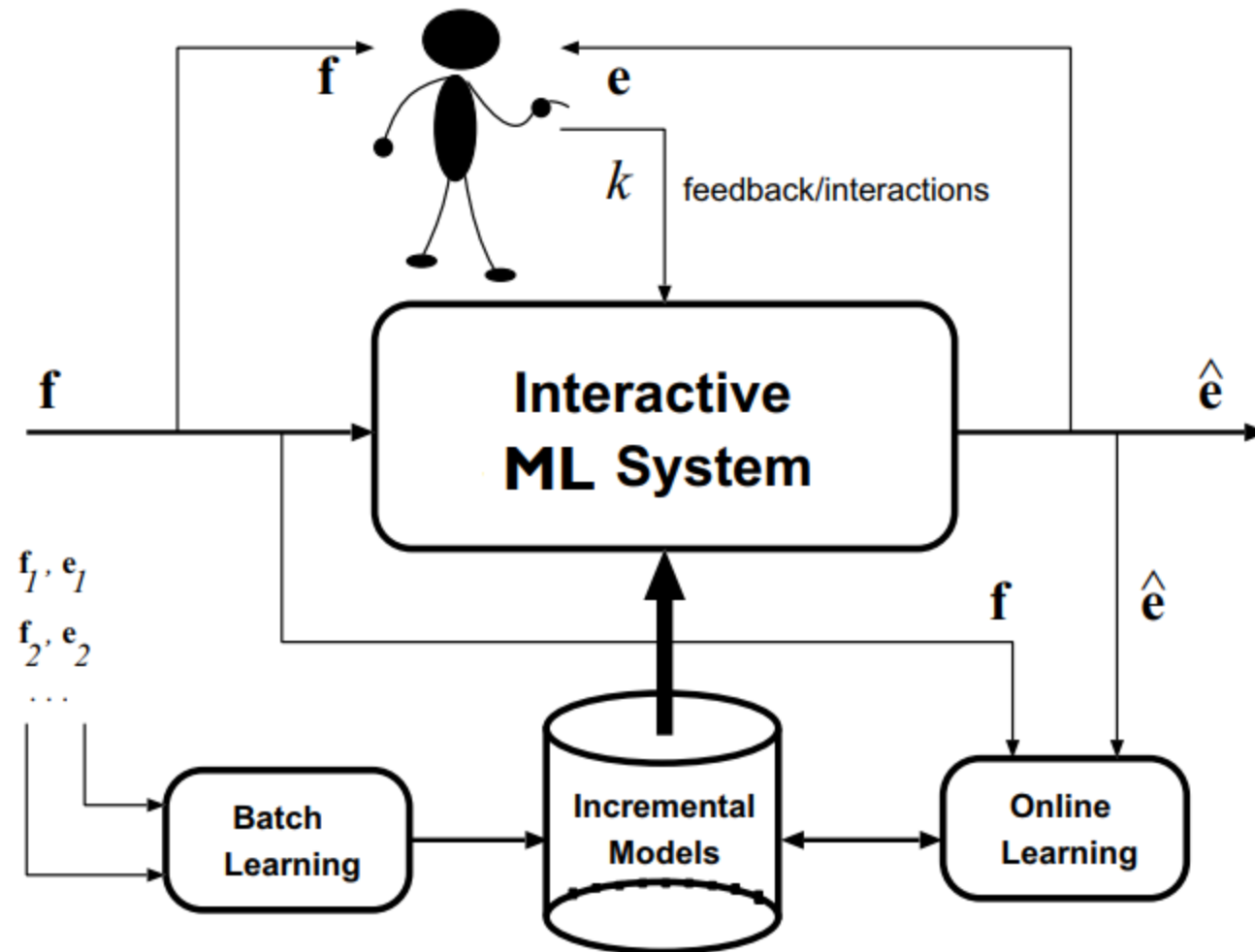
- Defining Characteristics
 - Expensive domain experts
 - Skewed class distribution(minority events)
 - Concept/ Feature drift
 - Biased sampling of labeled historical data
 - Lots of unlabeled data

Interactive Classification

Goal: Optimize life-time Return On investment



Another Scenario (Interactive ML)

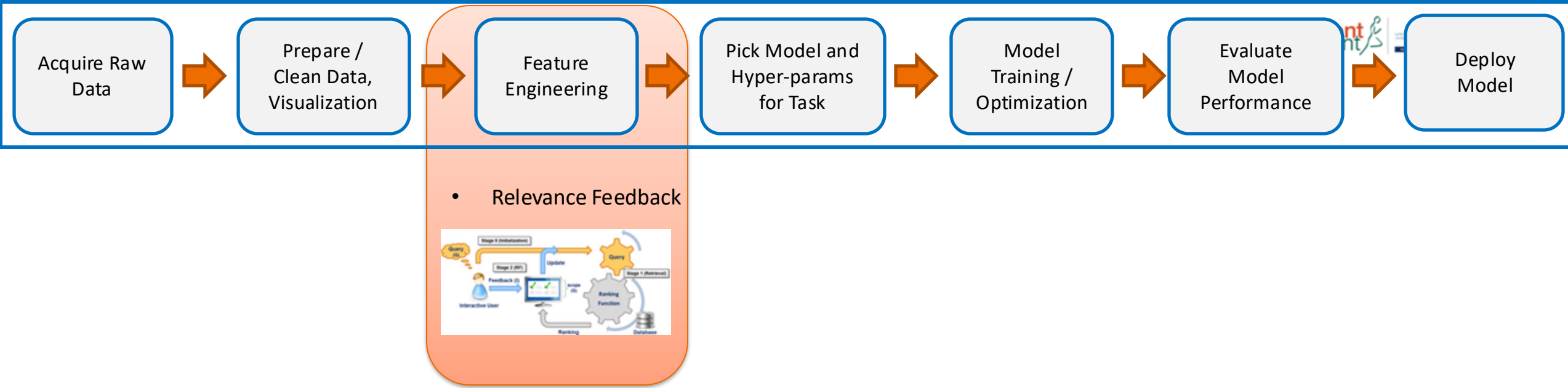


User may adjust:
- K (control) or
- Directly give desired e

Associated Issues

- Incremental and Computational Issues
 - How do we learn, adapt and forget
 - What is the basic knowledge and what do we adapt?
- Stability
 - Am I overlearning and changing too fast?
 - Stability, convergence and other algorithmic issues.

**Q: THREE EXAMPLES WHERE SYSTEM CAN
“SLOWLY” TAKE OVER FROM HUMANS**



Relevance Feedback

(Interactive Classification)

User Feedback

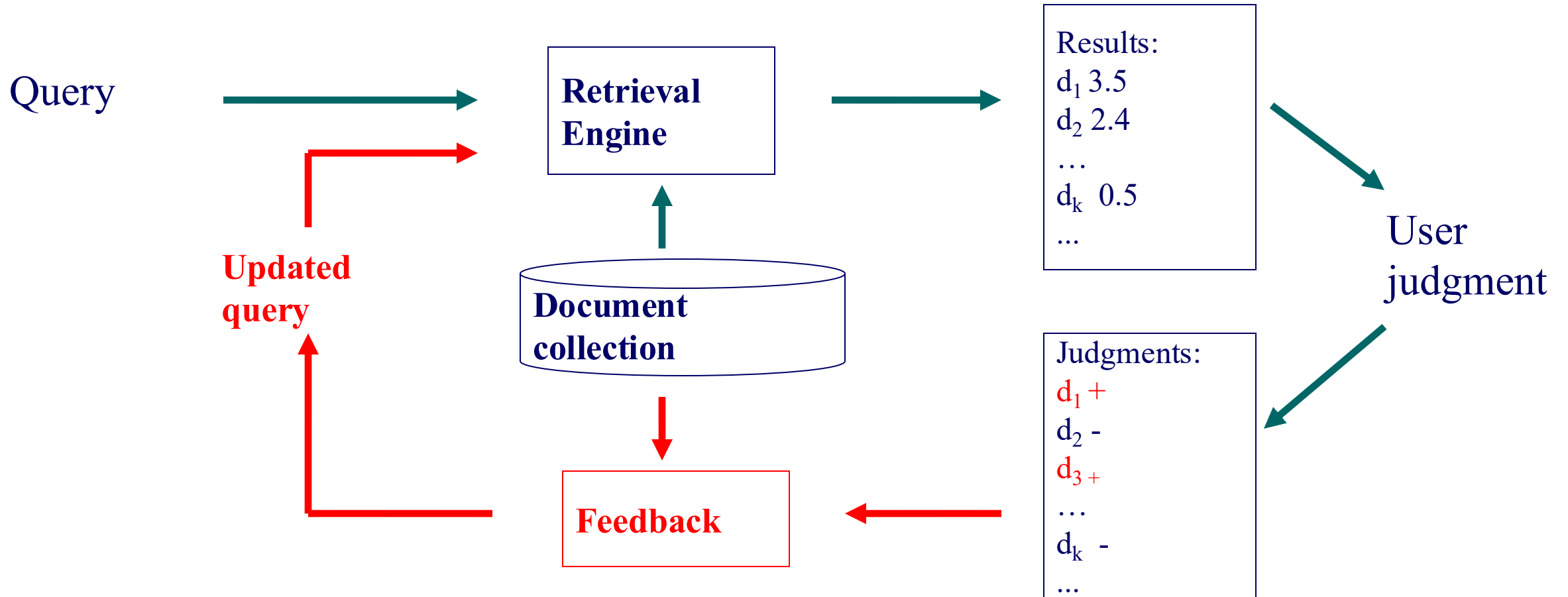
- An IR system could be an interactive system



Use Scenario

- A query q or a classifier w is given.
- Search engine retrieves a set of possible answers.
 - x_1, x_2, x_3 , etc.
- System guess the user intend and improve the answers.
 - x_7, x_{12}, x_{23} , etc.
- User is able to smartly navigate and get what she is looking for.
- E.g. Search for a specific fashion/design in a large database.

Relevance feedback



Effective and Popular(?)

[Personalization - Wikipedia, the free encyclopedia](#) 

Personalization involves using technology to accommodate the differences between individuals. Once confined mainly to the Web, it is increasingly becoming a ...

[en.wikipedia.org/wiki/Personalized](#) - 42k - [Cached](#) - [Similar pages](#) - 

Relevant

[Personalized Gifts from Personalization Mall](#) 

It shows you went out of your way to find the perfect gift at **to personalize** it to make it theirs alone! At PersonalizationMall.com, we design most of our ...

[www.personalizationmall.com/Default.aspx?&did=111028](#) - 47k -

[Cached](#) - [Similar pages](#) - 

Nonrelevant

[What is personalization? - a definition from Whatis.com](#) 

Mar 6, 2007 ... On a Web site, **personalization** is the process of tailoring pages to individual users' characteristics or preferences.

[searchcrm.techtarget.com/sDefinition/0,,sid11_gc532341,00.html](#) - 72k -

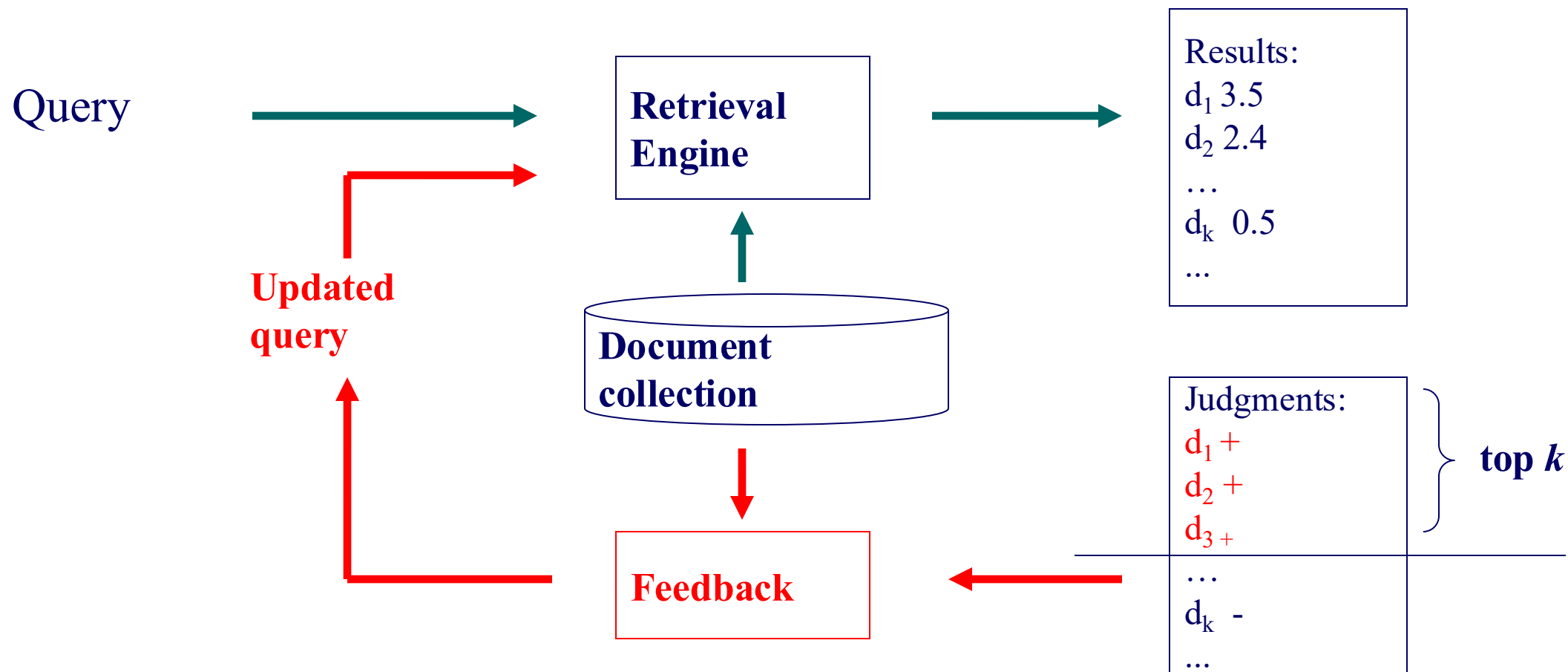
[Cached](#) - [Similar pages](#) - 



Too Explicit?

Pseudo feedback and Query Expansion

- What if the users are reluctant to provide any feedback



Rocchio Model

$$Q_1 = \alpha Q_0 + \frac{\beta}{n_1} \sum_{i=1}^{n_1} R_i - \frac{\gamma}{n_2} \sum_{i=1}^{n_2} S_i$$

where

Q_0 = the vector for the initial query

R_i = the vector for the relevant document i

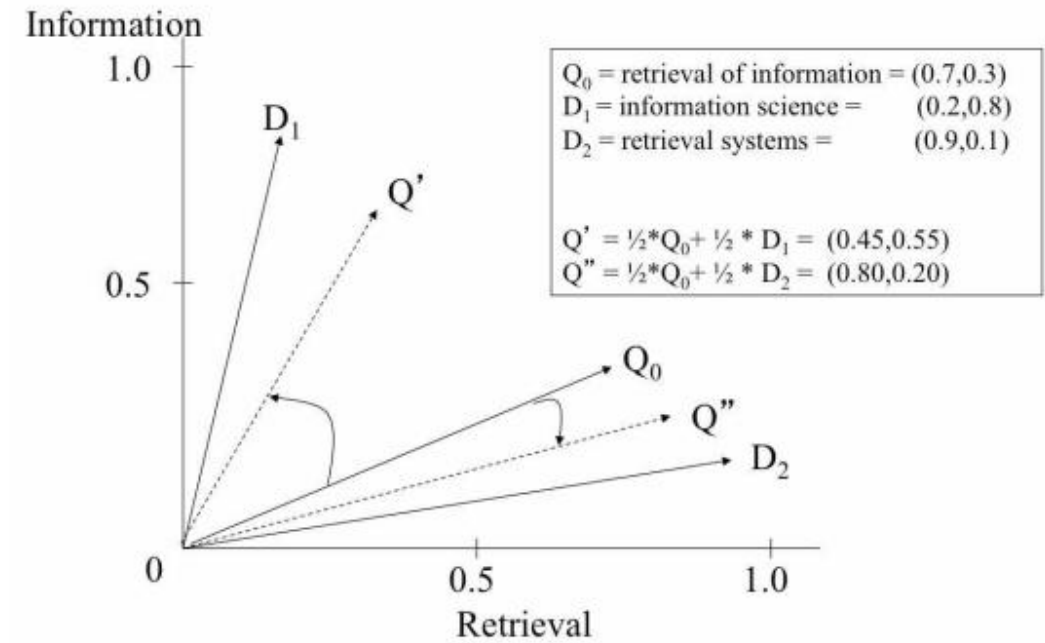
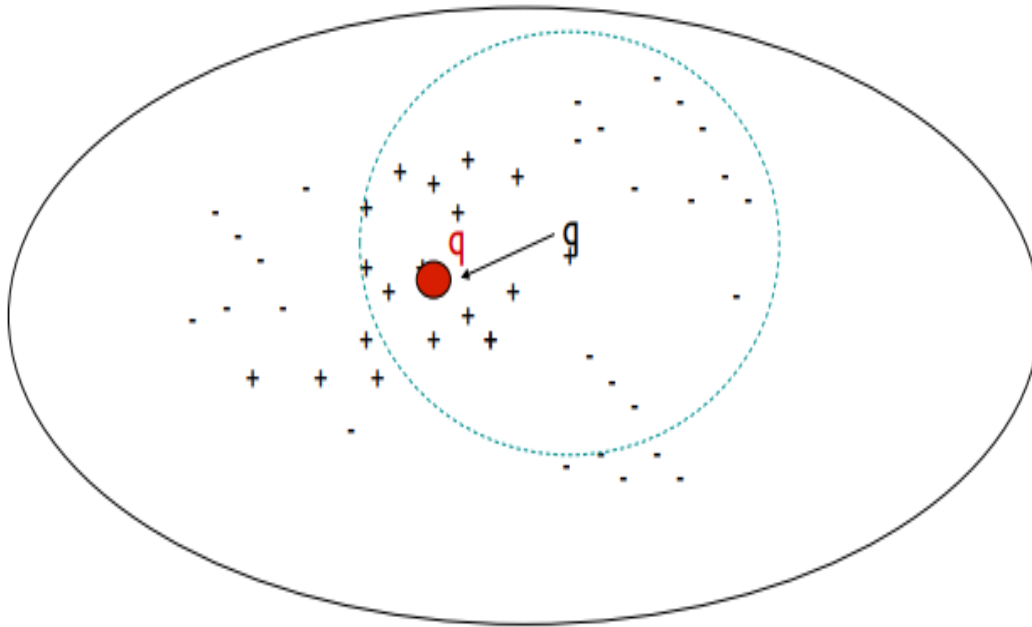
S_i = the vector for the non - relevant document i

n_1 = the number of relevant documents chosen

n_2 = the number of non - relevant documents chosen

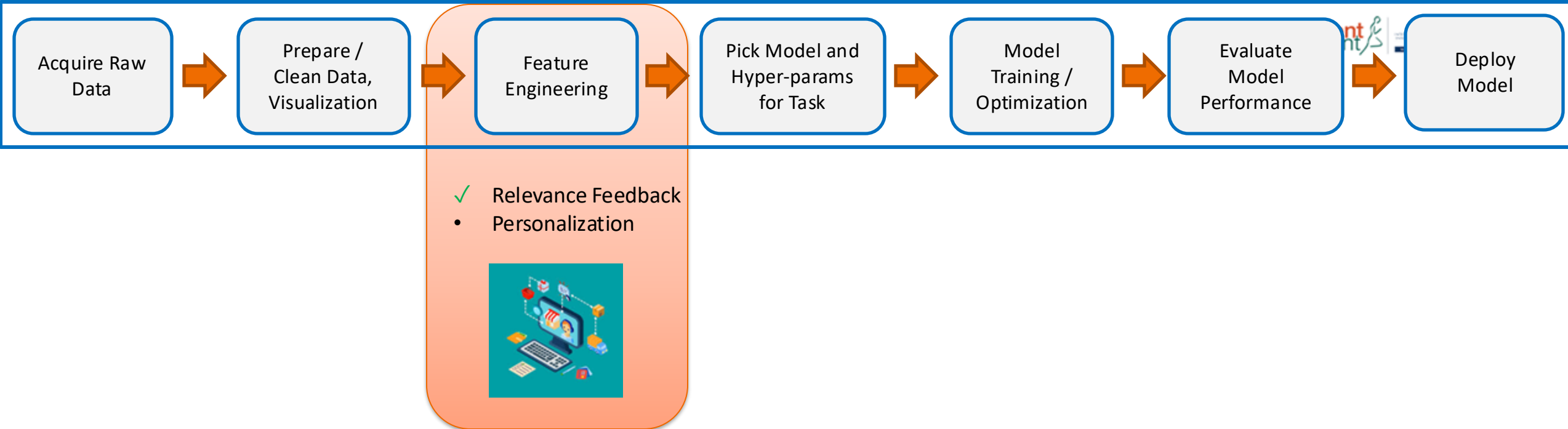
α , β and γ tune the importance of relevant and nonrelevant terms (in some studies best to set β to 0.75 and γ to 0.25)

Illustration



Challenges and Refinements

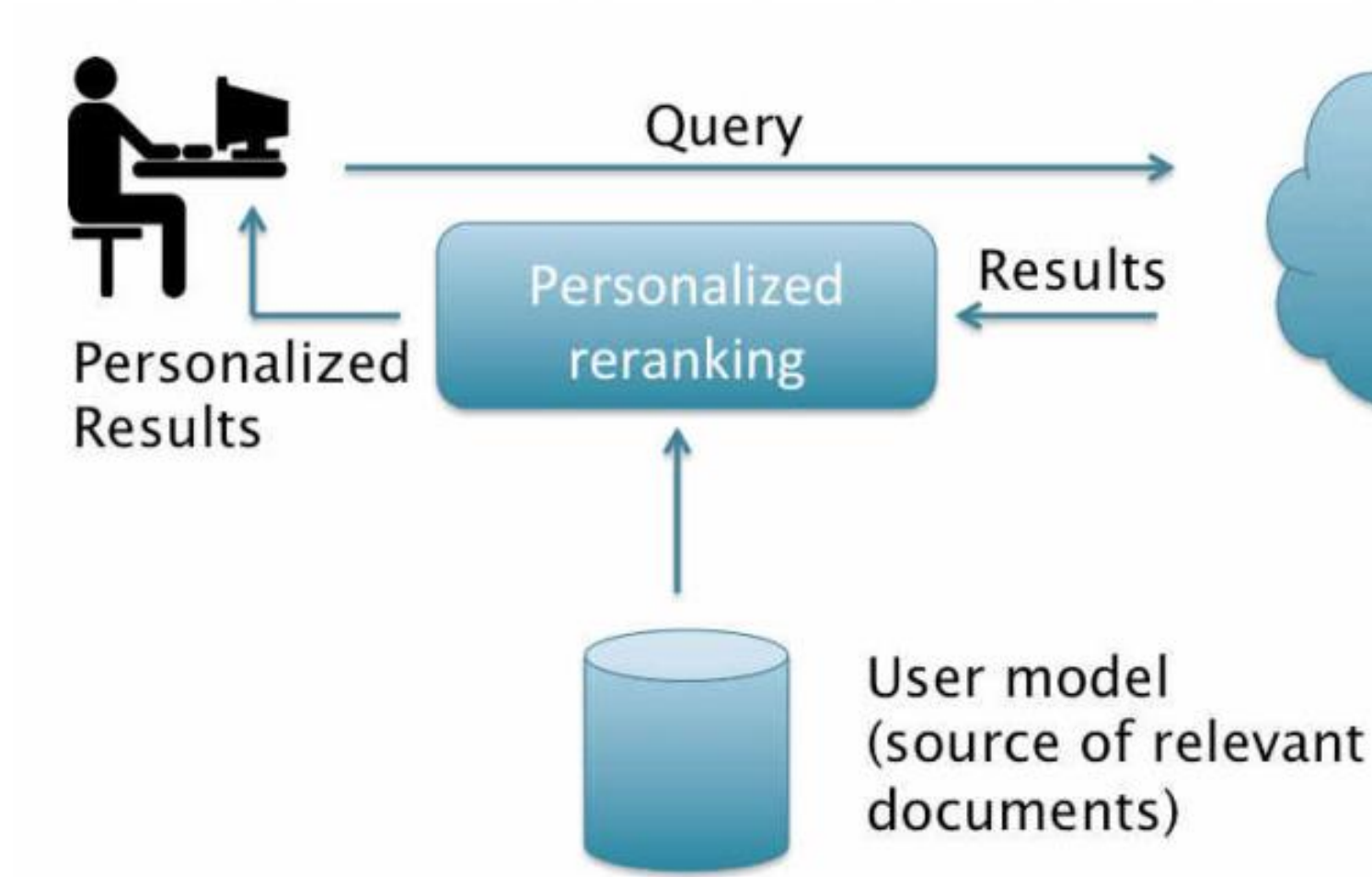
- Can we force user to say + and – on the answers?
 - Often + is more clear ?. But not –ve is not shared.
 - Cases when only + or Only – is available.
- Often + is implicit (I click/browse) and not explicit.
- **Examples:**
 - Browsing for fashion (clothes)



Personalization



Use Case (RF style)



Many Relevant Attributes

- Consider the following pieces of information
 - Geographical Location
 - Age, gender, ethnicity, religion, etc.
 - Interests
 - Previous reviews on products
 -
- How could these pieces of information help?
- How to collect these information?

Approaches

- Individual Vs Collaborative
- Reactive Vs Proactive
- User Vs Item Information

Individual Vs Collaborative

- Individual approach (E.g. Google Personalized Search)
 - Use only individual user's data
 - Generate user profile by analyzing
 - User's browsing behavior
 - User's active feedback on the system
- Advantage
 - Can be implemented on the client-side - no privacy violation
- Disadvantage
 - Based only on past interactions.

Reactive Vs Proactive

- Reactive approach
 - Explicitly ask user for preferences
 - Either in the form of query or feedback
- Proactive approach
 - Learn user preferences by user behavior
 - No explicit preference demand from the user
- Behavior is extracted
 - Click-through rates
 - Navigational pattern

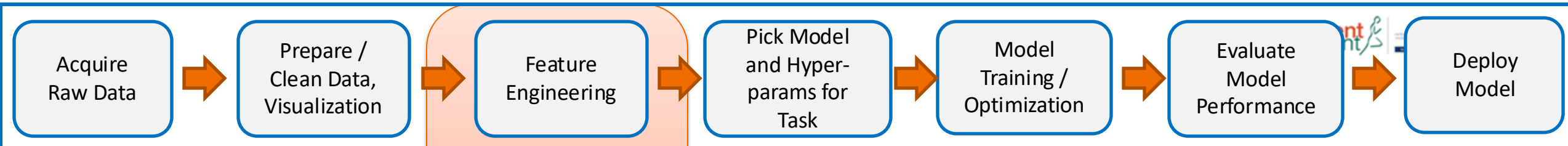
User Vs Item Information

- User Information
 - Geographic location (from IP address)
 - age, gender, marital status, etc. (explicit query)
 - Lifestyle, etc. (inference from past behavior)
- Item Information
 - Content of Topics – movie genre, etc.
 - Product/ domain ontology

Summary

- ML systems can work with human
 - While Training
 - While Testing
 - To Learn, To Help
- A strategy to deploy and slowly improve
- A strategy to co-exists with humans
 - From Indian view point
 - From global view point

Questions?



- ✓ Relevance Feedback
- ✓ Personalization
- Beyond the Full Supervision

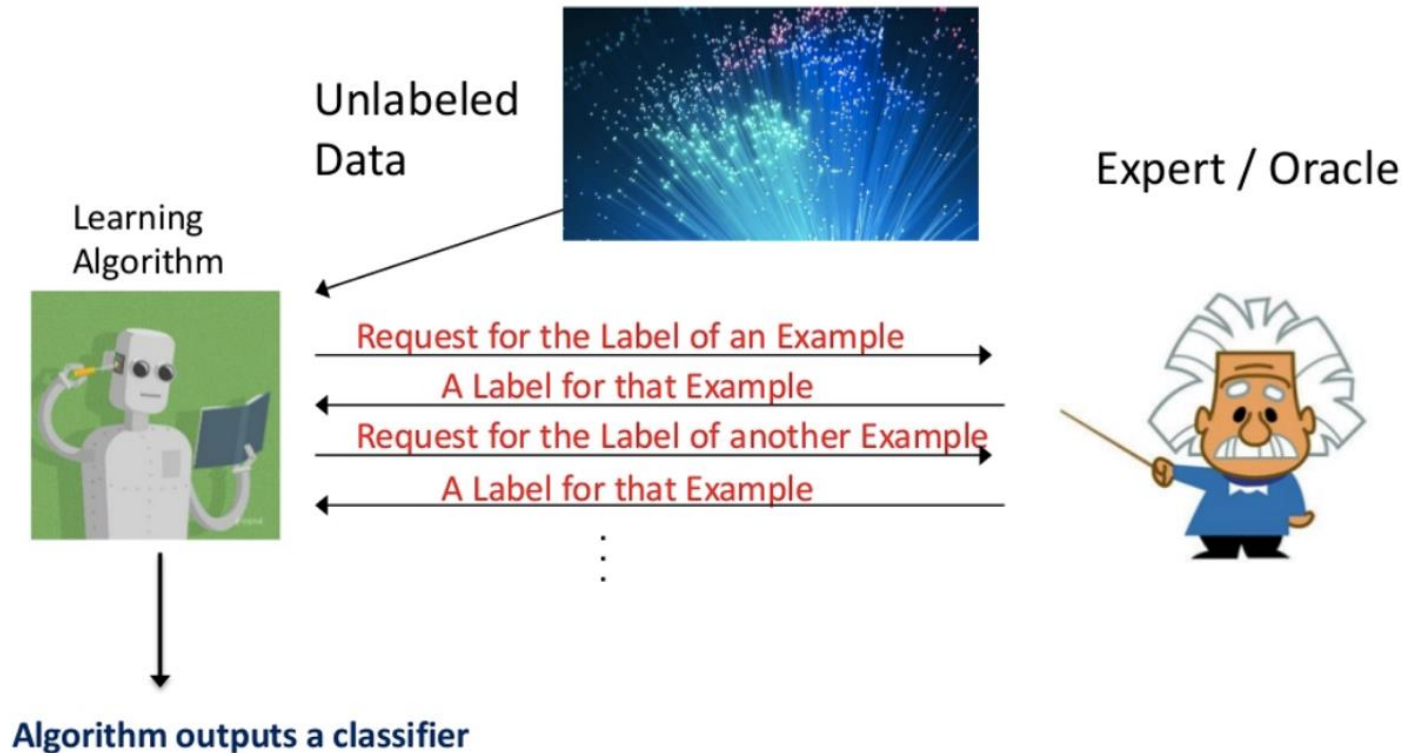


Beyond the Full Supervision

Challenges with Supervision

- I have too much data. But most of them are unlabeled. What do we do?
- I have labeled data. But a good percentage of the labels are erroneous. What do I do?
- I have labeling's from experts itself. But they do not agree. What do we do?
- My supervisors are too costly. How do I do minimize the cost of supervision?
- ...

Learn with minimal # of examples ?



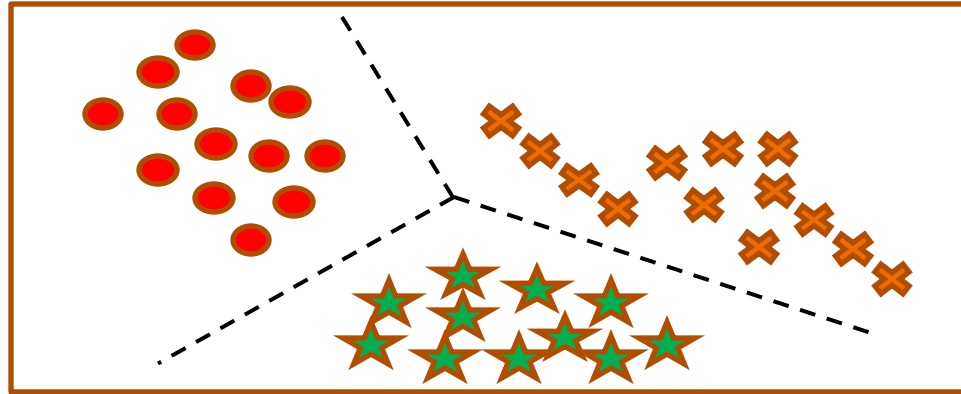
Active Learning

- Eg. Learn the notion of a rectangle.

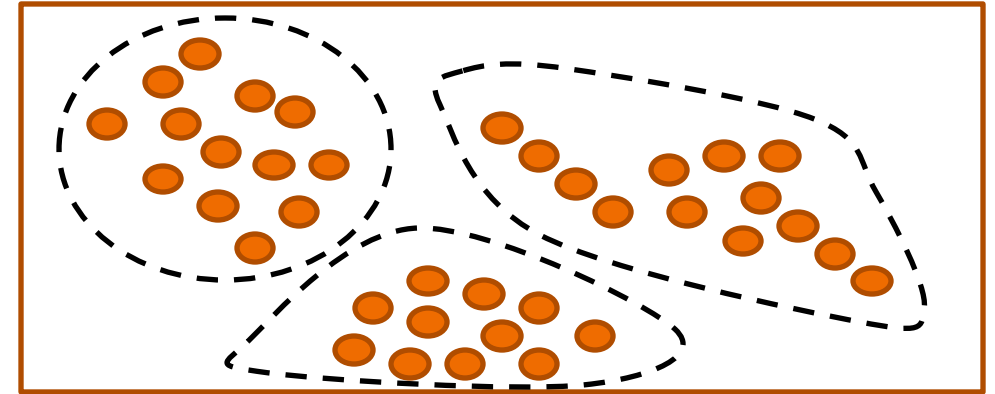
Semi Supervised Learning

- I have a small quantity of labelled data and large quantity of unlabeled data.
 - How do I take advantage of the unlabeled data?

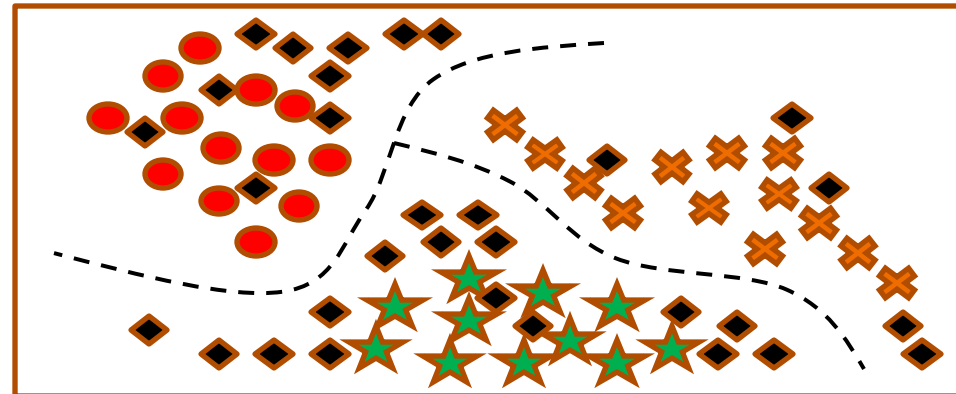
Algorithms



Supervised learning



Unsupervised learning



Semi-supervised learning

Self Training: Naïve

- Train a supervised learner on available labelled data (X_l, Y_l) .
- Label all points in unlabeled data X_u .
- Retrain the classifier using the new labels for documents where the classifier is most confident.
- Continue until labels do not change any more.

Self Training: Refined

- **Assumption:** One's own high confidence predictions are correct.
- **Self-Training Algorithm**
 - Train on labeled examples
 - Predict on unlabeled examples
 - Add $(x, f(x))$ to the labeled data
 - Add all
 - Add a few most confident pairs
 - Add weight for each pairs
 - Repeat the process

Co-Training

- Co-training assumed two “Views” of the data where each input x is a pair

$$x = (x_1, x_2)$$

- Eg. In the context of web page classification,
 - x_1 may be metadata associated with the web page such as title etc.
 - x_2 be the words in the link pointing to this page.
- Assume there exists functions c_1 , c_2 and c such that

$$c_1(x_1) = c_2(x_2) = c(x)$$

- Two sets of features x_1 and x_2 are conditionally independent given the class.

1998 paper demonstrates,
with 12 labeled examples, 788
web pages could be classified
with 95% accuracy.

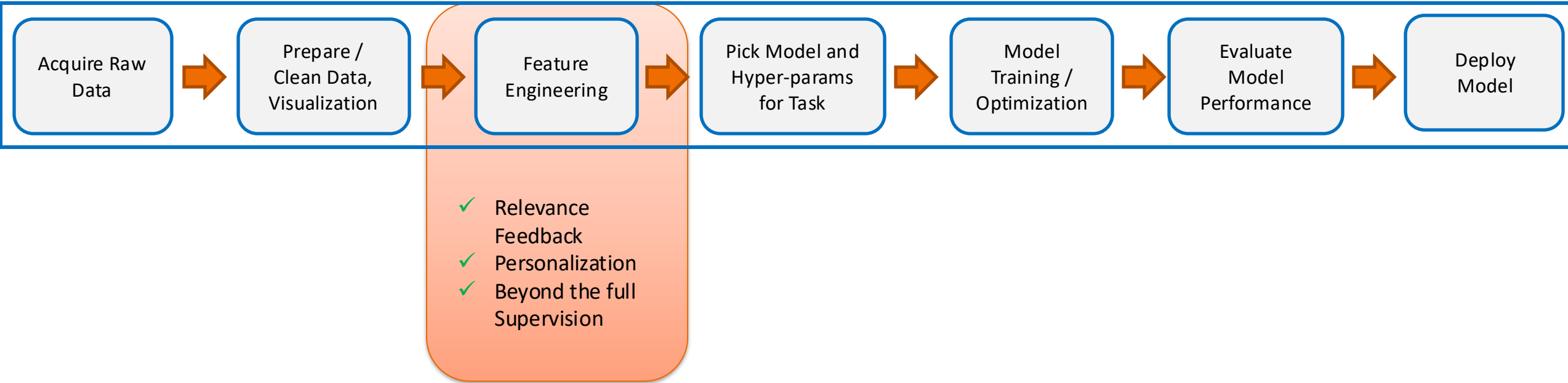
Co-Training

- Use the labeled data to learn the initial h_1, h_2
- First use h_1 to label examples that it is confident about and then feed these to our learner to update h_2
- Then use h_2 to label examples that it is confident about and then feed these to our learner to update h_1
- Keep repeating this process

Summary: Questions?

- Varying amount and quality of supervision
 - Many wrapper style methods.
 - Intuitive
- Many principled formulations
 - Formal extensions of existing methods
 - (eg. Transductive SVMs; Semi Supervised Random Forest)
 - Many newer learning problems
 - (eg. Multiple Instance Learning,)

Summary



Thanks!!

Questions?